

AI-03: Homestay Guest Review Sentiment Classifier

Project Overview

Trishul Eco-Homestays receives guest reviews from multiple platforms such as Google, Booking.com, TripAdvisor, and Instagram. Currently, there is no automated system to analyze these reviews, making it difficult for management to identify customer satisfaction trends, recurring issues, and areas for improvement.

This project aims to develop an AI-powered web application that can automatically classify guest reviews, identify the main topic of feedback, and generate appropriate management responses.

Problem Statement

Guest reviews are spread across different online platforms and must be analyzed manually. This process is time-consuming and makes it difficult to quickly identify positive feedback, customer complaints, and service improvement opportunities.

Proposed Solution

Develop a web-based review classification tool where staff members can paste one or multiple guest reviews. The system will use an LLM API (Gemini API) to:

- Classify sentiment as Positive, Neutral, or Negative.
- Identify the primary theme of the review.
- Generate a suitable management response.
- Display results in a structured table.

Objectives

1. Automate guest review analysis.
2. Reduce manual effort required for review categorization.
3. Help management identify recurring customer concerns.
4. Improve customer engagement through suggested responses.
5. Demonstrate practical use of AI in hospitality management.

Target Users

- Homestay Owners
- Property Managers
- Customer Support Teams
- Hospitality Businesses

Technology Stack

- React.js
- HTML
- CSS
- JavaScript
- Gemini API
- GitHub

Key Features

- Single and bulk review analysis.
- Sentiment classification.
- Theme detection (Food, Host, Location, Cleanliness, Value, Experience).
- AI-generated management response.
- User-friendly dashboard with tabular results.

Expected Deliverables

1. Functional web application.
2. Gemini API integration.
3. Sentiment classification module.
4. Theme tagging module.
5. Suggested response generator.
6. Test report based on 20 real or simulated reviews.
7. GitHub repository with source code and documentation.

Expected Impact

The proposed solution will help Trishul Eco-Homestays efficiently analyze customer feedback, improve service quality, and make data-driven decisions to enhance guest satisfaction.